

HYDRATE LESS, RENDER MORE WITH NUXT

Vue.js Hamburg Meetup

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Senior Full Stack Developer



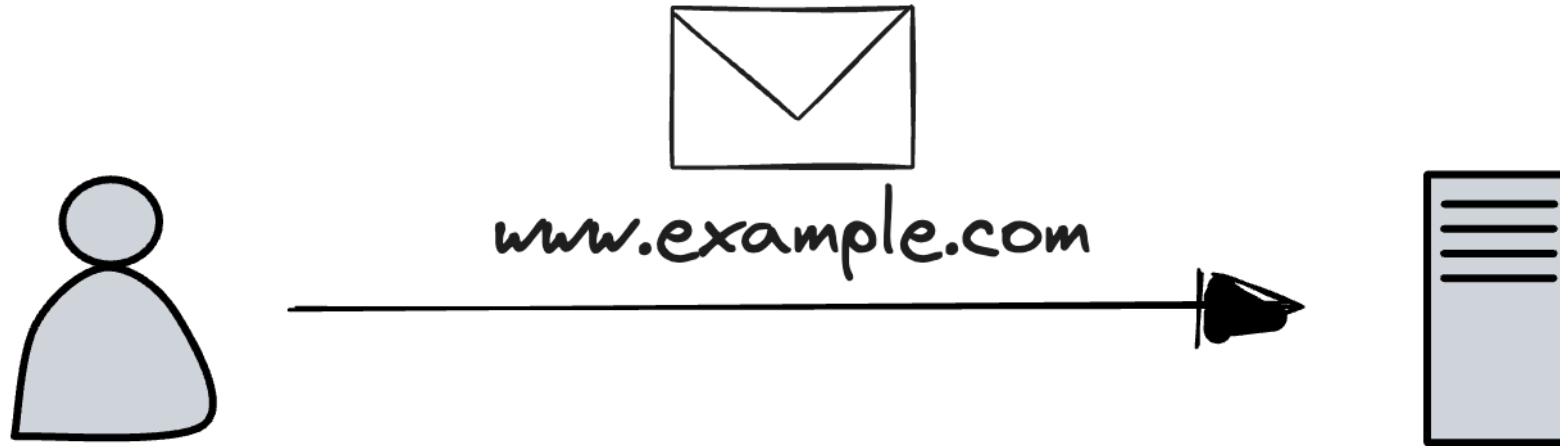
- 5 years of Vue/Nuxt experience

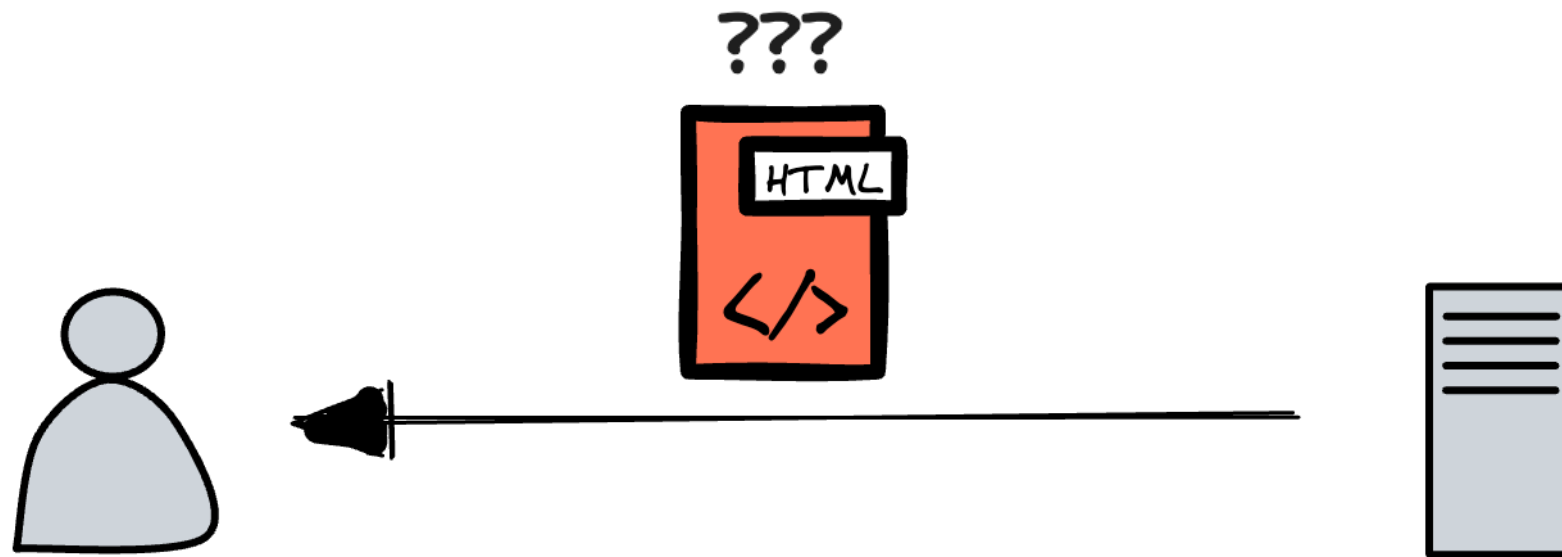


RENDERING MADNESS

- ~20 years ago
 - Me giving a presentation about PHP
 - Working with pure HTML
- Now
 - SSR, CSR, SSG, Hydration, Lazy-Hydration
 - Islands







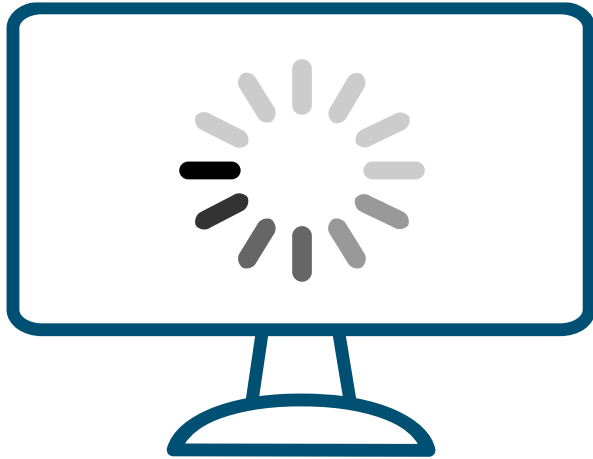
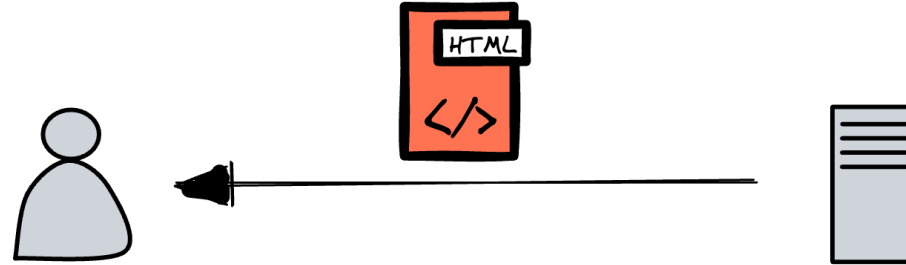


```
CSR example

<!DOCTYPE html>
<html>
  <head>
    <title>Nuxt Rendering Modes Demo</title>
    <script type="module" src="/_nuxt/B_5vHM_Q.js" crossorigin></script>
  </head>
  <body>
    <div id="__nuxt"></div>
  </body>
</html>
```

```
SSR example

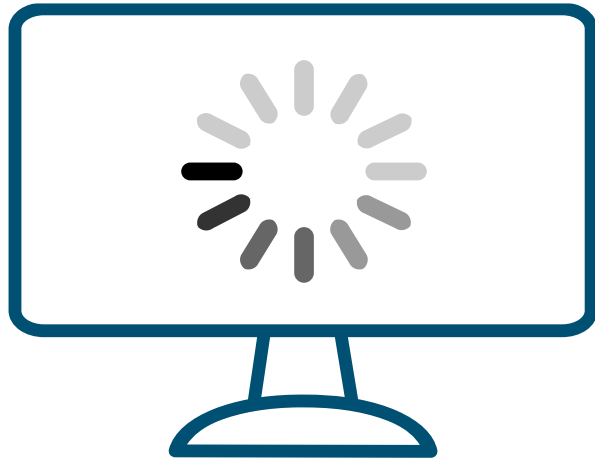
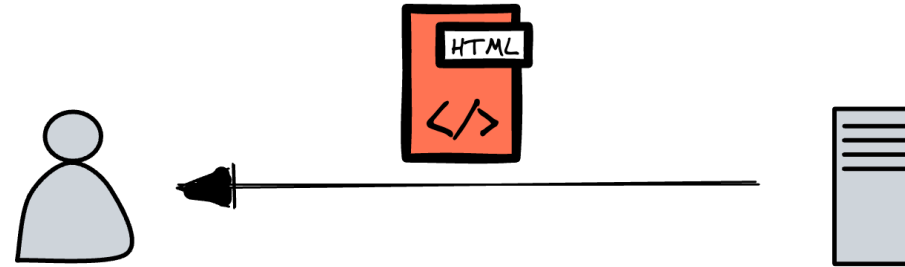
<!DOCTYPE html>
<html>
  <head>
    <title>Nuxt Rendering Modes Demo</title>
    <script type="module" src="/_nuxt/B_5vHM_Q.js" crossorigin></script>
  </head>
  <body>
    <div id="__nuxt">
      <div>
        <header class="header">
          <div class="header-content">
            <h1 class="logo">
              <a href="/">🚀 Nuxt Render Modes Demo</a>
            </h1>
            <nav class="nav">
              <a href="/" class="nav-link">Home</a>
              <a href="/ssr" class="active nav-link">SSR</a>
              ...
            </nav>
          </div>
        </header>
        <main class="container">
          ...
        </main>
      </div>
    </div>
  </body>
</html>
```



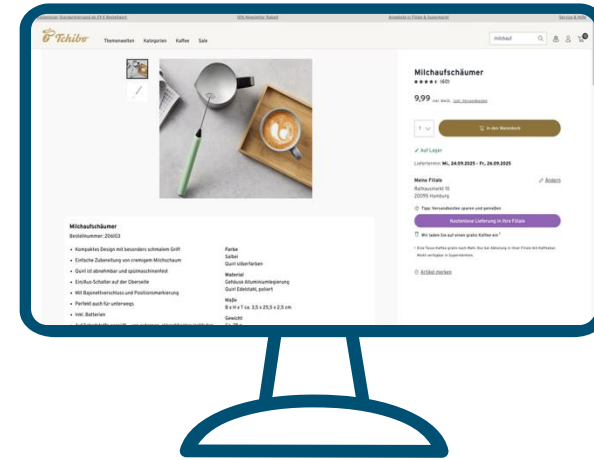
Client-side rendering (CSR)

```
SSR example

<!DOCTYPE html>
<html>
  <head>
    <title>Nuxt Rendering Modes Demo</title>
    <script type="module" src="/_nuxt/B_5vHM_Q.js" crossorigin></script>
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  </body>
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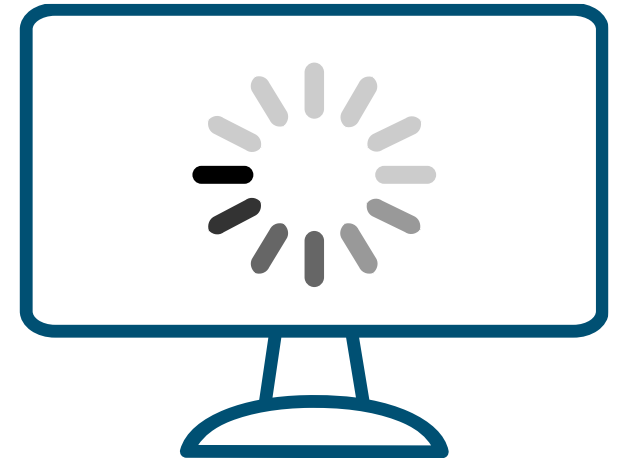
Client-side rendering (CSR)



Server-side rendering (SSR)

CLIENT-SIDE RENDERING (CSR)

- Renders content only in browser
- Highly interactive pages
- Downsides
 - Initial load takes some time
 - HTML content not included in first response

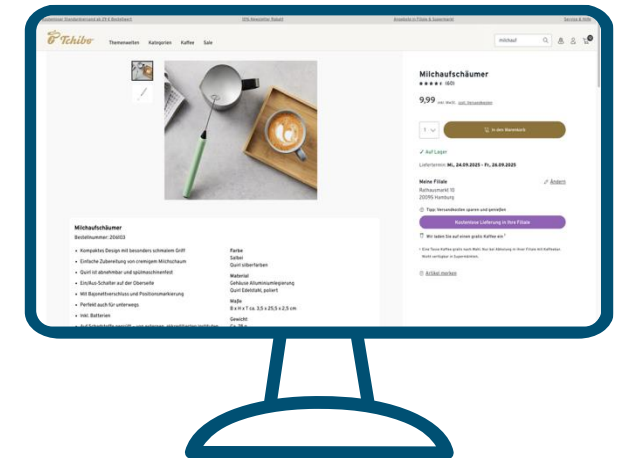


STATIC SITE GENERATION (SSG)

- Rendering at build time instead of per request (SSR)
- Suitable for less frequently changed content
- Can contain interactive content

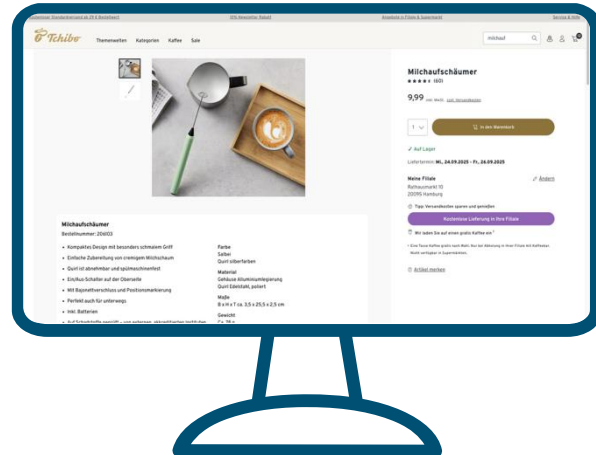
SERVER-SIDE RENDERING (SSR)

- Page is rendered on the server-side
- HTML content delivered in first response
- Fast page load times
- Hydration phase for interactivity

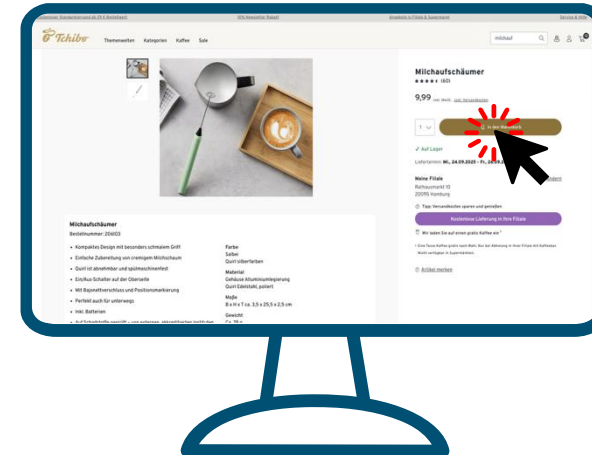


HYDRATION

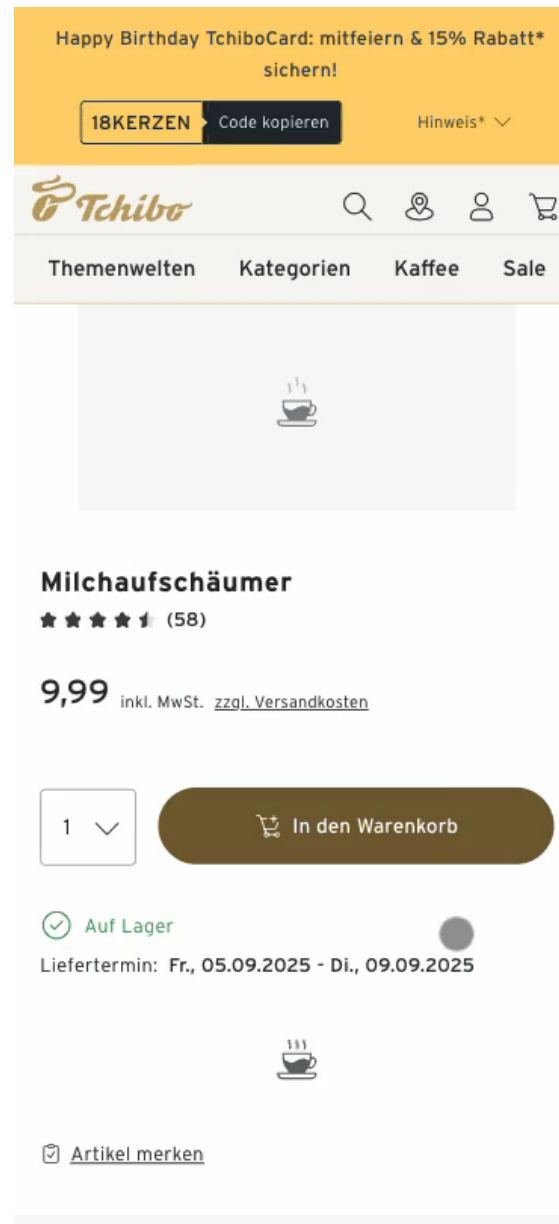
1. Match HTML elements to Vue components
2. Attach event listeners to HTML elements



Static



Interactive



Test settings:

6x CPU slow-down

Slow 4G network

RAGE CLICK DEMO

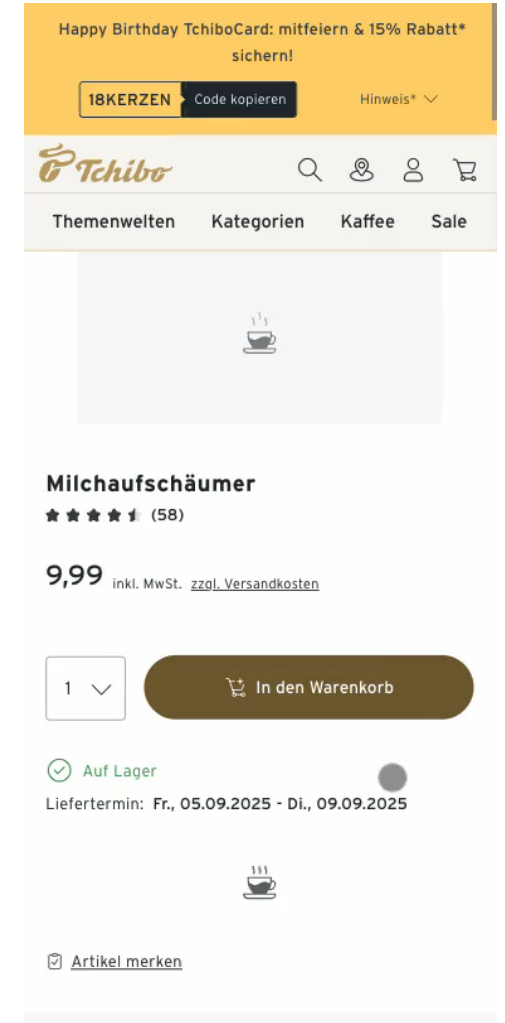
- Problem of every hydrated application
- Hydration time can't be eliminated
- Strategies



No hydration



Delay hydration



ISLANDS

- Static content sprinkled with interactive islands
- Hydration prioritization of islands
- Reduced likelihood for users to experience rage clicks

NUXT ISLANDS

- Interactive content with mostly non-interactive islands
 - Can still be interactive using props with one caveat
- Content rendered fully on the server-side
- Reduced hydration time, smaller bundle size



NUXT LAZY HYDRATION

- Not a rendering mode, but an optimization technique
- Defer hydration of non-critical interactive components
- Can be applied to SSR, SSG

DEMO TIME



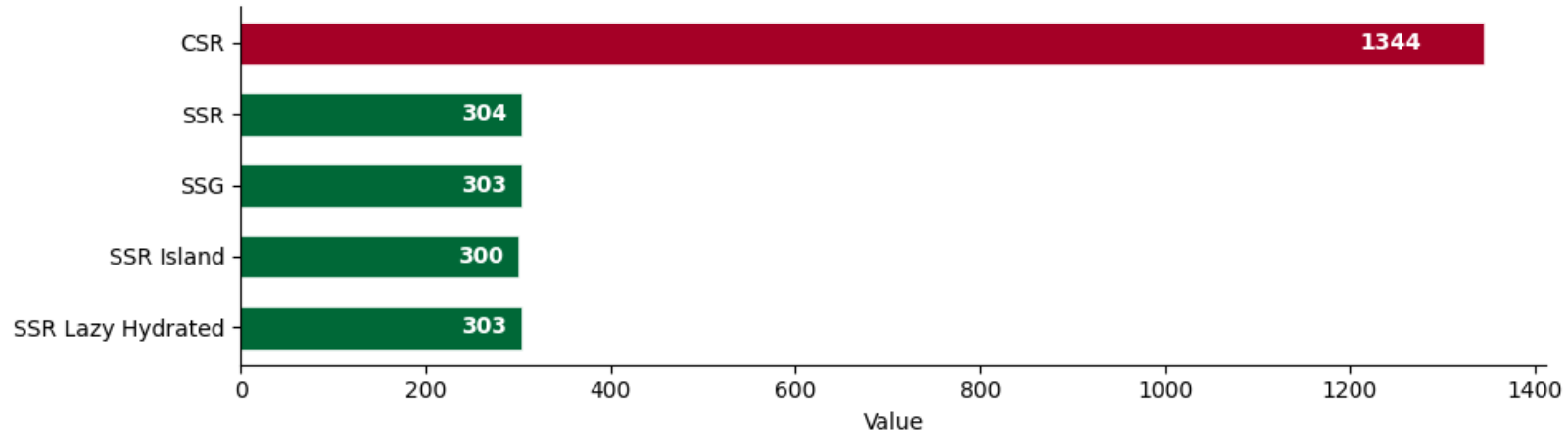
<https://blog.lancedb.com/guide-to-use-contextual-retrieval-and-prompt-caching-with-lancedb/>

COMPARISON OF RENDER MODES

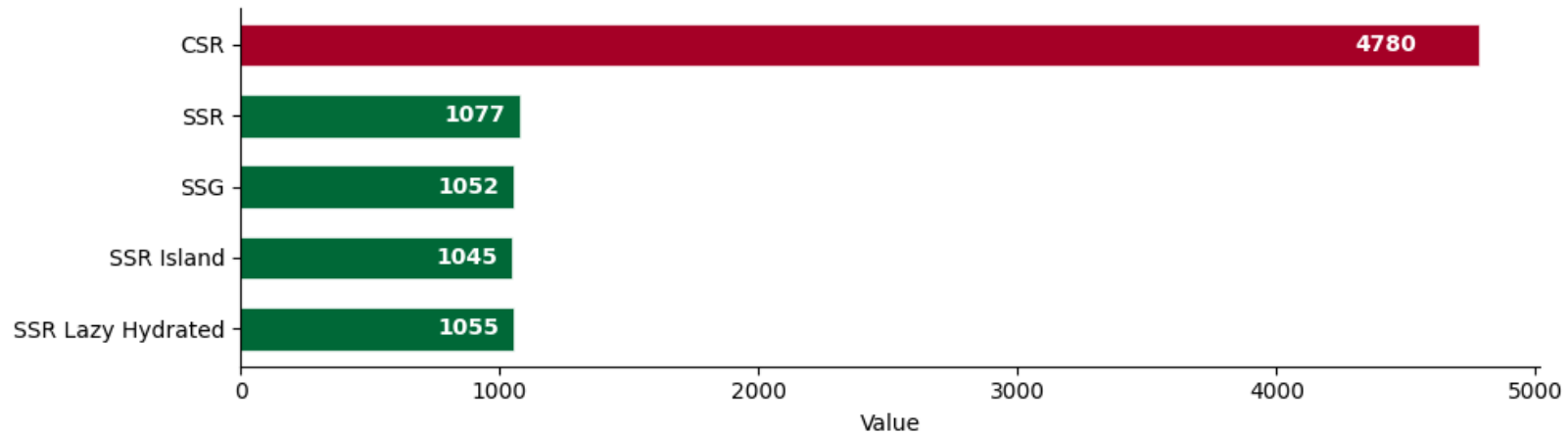
- Demo Nuxt 3 application
- Page load test (10 runs, p75, medium/high throttling)
- Criteria
 - First Contentful Paint (FCP)
 - Hydration Time
 - Client Bundle Size

FCP – p75, 10 runs (ms)

Medium throttling

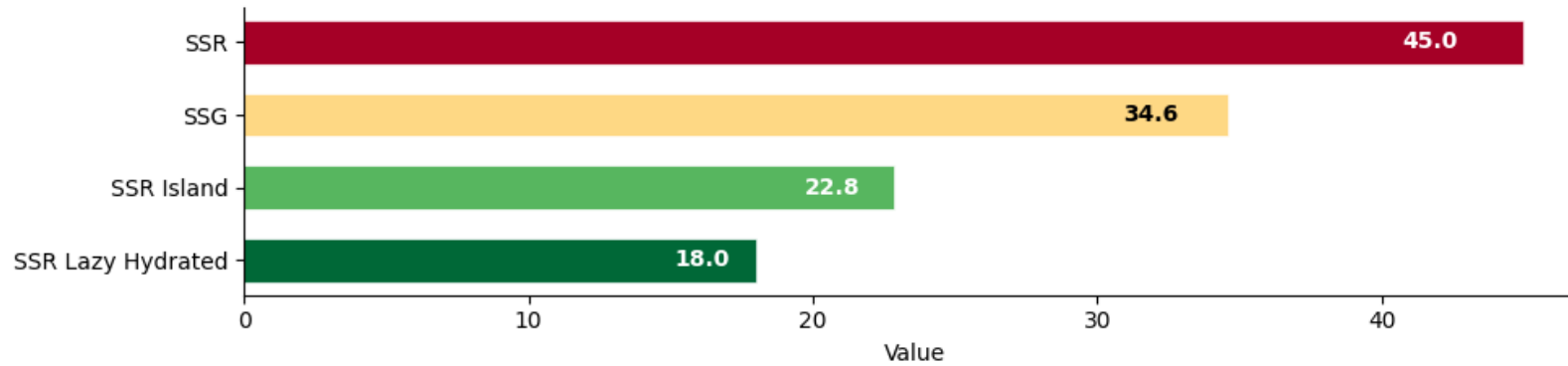


High throttling

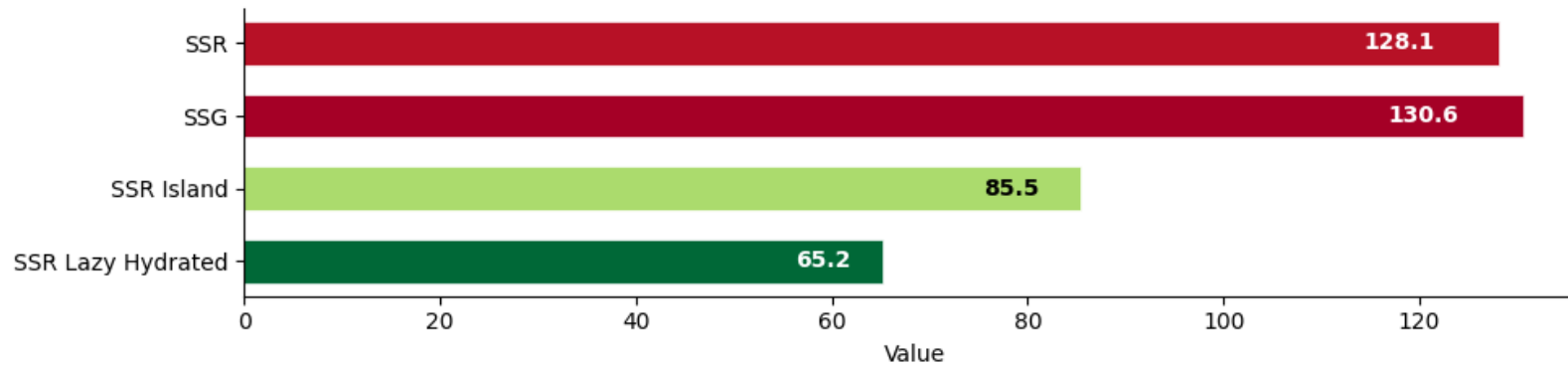


Hydration Time – p75, 10 runs (ms)

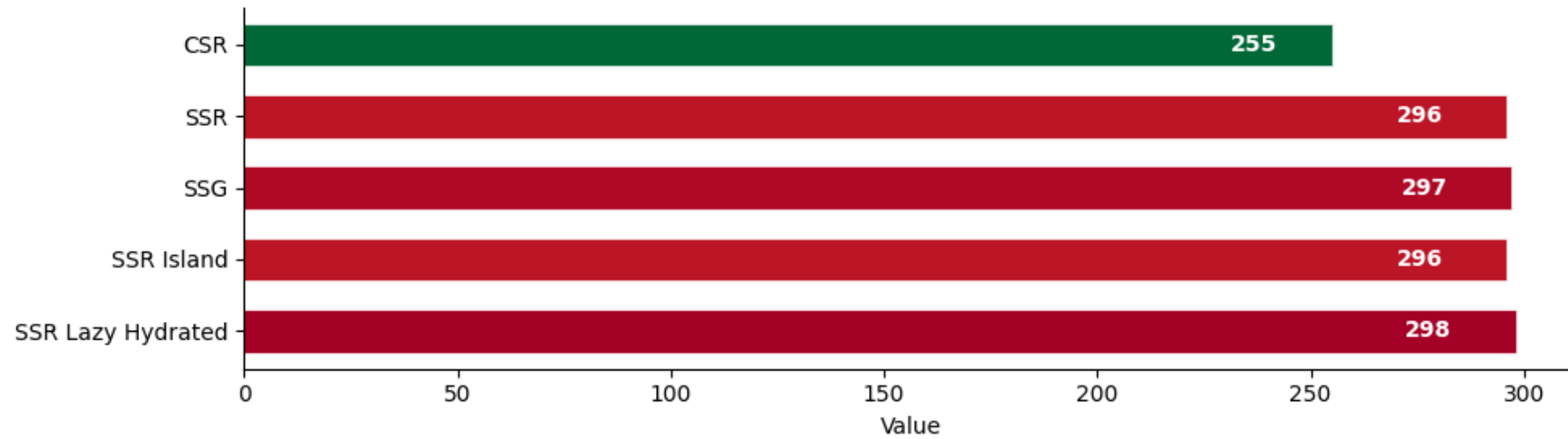
Medium throttling



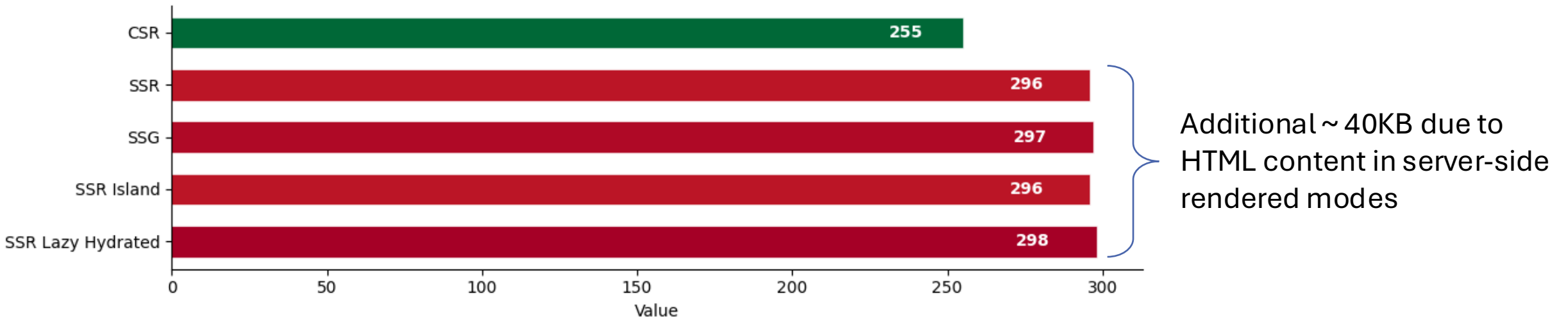
High throttling



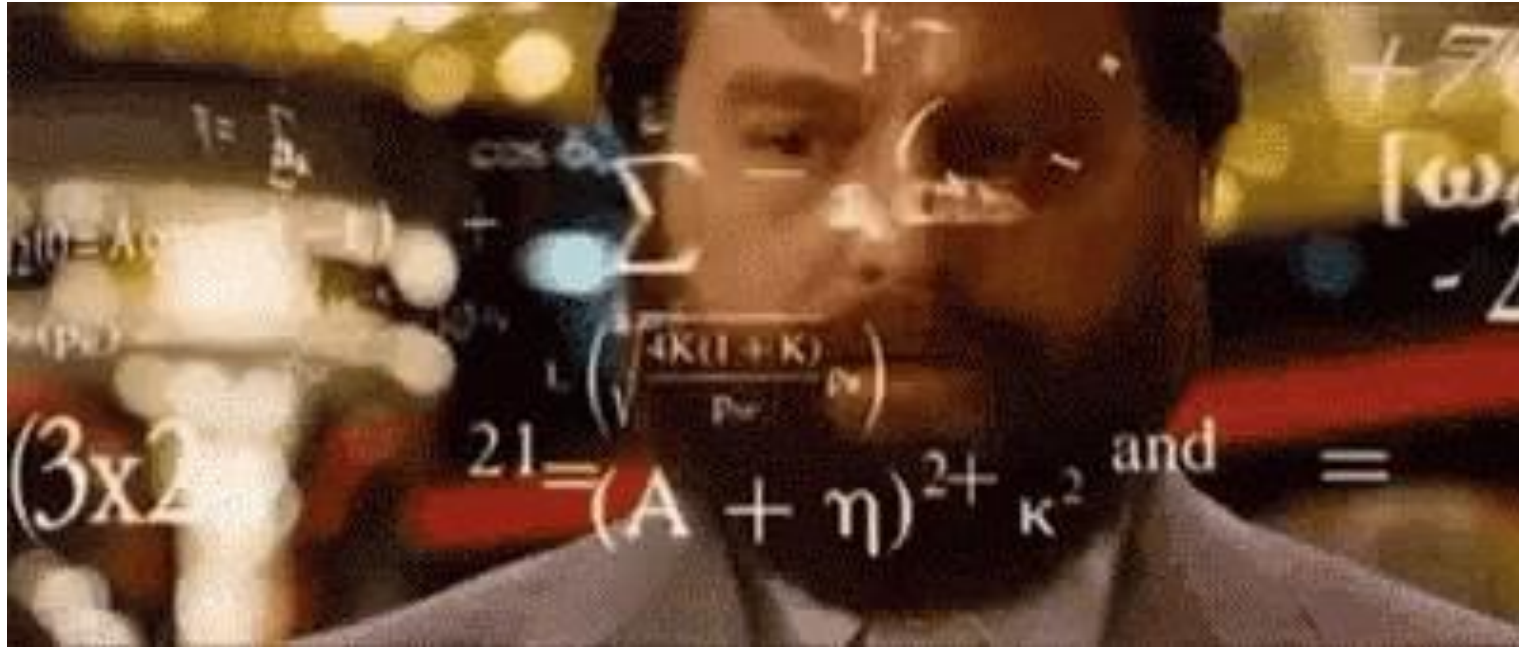
Bundle Size (KB)



Bundle Size (KB)



WHICH IS THE RIGHT RENDERING MODE FOR MY APP?



PICK THE RIGHT RENDERING MODE

Criteria	CSR	SSR	SSG	Nuxt Islands

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Criteria	CSR	SSR	SSG	Nuxt Islands
SEO	✗	✓	✓	✓
Fast FCP	✗	✓	✓	✓

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Criteria	CSR	SSR	SSG	Nuxt Islands
SEO	✗	✓	✓	✓
Fast FCP	✗	✓	✓	✓
Interactive/personalized content	★	✓	✗	✗

PICK THE RIGHT RENDERING MODE

Criteria	CSR	SSR	SSG	Nuxt Islands
SEO	✗	✓	✓	✓
Fast FCP	✗	✓	✓	✓
Interactive/personalized content	★	✓	✗	✗
Mostly static with some interactivity	✗	✓	★	★

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Criteria	CSR	SSR	SSG	Nuxt Islands
SEO	✗	✓	✓	✓
Fast FCP	✗	✓	✓	✓
Interactive/personalized content	★	✓	✗	✗
Mostly static with some interactivity	✗	✓	★	★
Mostly interactive with some static parts	✓	✓	✓	★

PICK THE RIGHT RENDERING MODE

Criteria	CSR	SSR	SSG	Nuxt Islands
SEO	✗	✓	✓	✓
Fast FCP	✗	✓	✓	✓
Interactive/personalized content	★	✓	✗	✗
Mostly static with some interactivity	✗	✓	★	★
Mostly interactive with some static parts	✓	✓	✓	★
Complex server logic/sensitive data	✗	✓	✗ (✓ at build time)	✓

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Criteria	CSR	SSR	SSG	Nuxt Islands
SEO	✗	✓	✓	✓
Fast FCP	✗	✓	✓	✓
Interactive/personalized content	★	✓	✗	✗
Mostly static with some interactivity	✗	✓	★	★
Mostly interactive with some static parts	✓	✓	✓	★
Complex server logic/sensitive data	✗	✓	✗ (✓ at build time)	✓
Client Bundle Size/Number of Requests	✓	✓	✓	★

Always Benchmark



Consider islands

Identify lazy-loadable parts

What is the purpose of the page?

THANK YOU!

*Check out the
code here:*

